



SMART INDIA
HACKATHON
2022

Basic Details of the Team and Problem Statement

Ministry/Organization Name/Student Innovation:

Ministry of Ports, Shipping & Waterways

PS Code: AK1137

Problem Statement Title: Air Quality Index for Major Port

Team Name: VAAYU

Team Leader Name: Manirajan K

Institute Code (AISHE): U-0447

Institute Name: BHARATHIAR UNIVERSITY

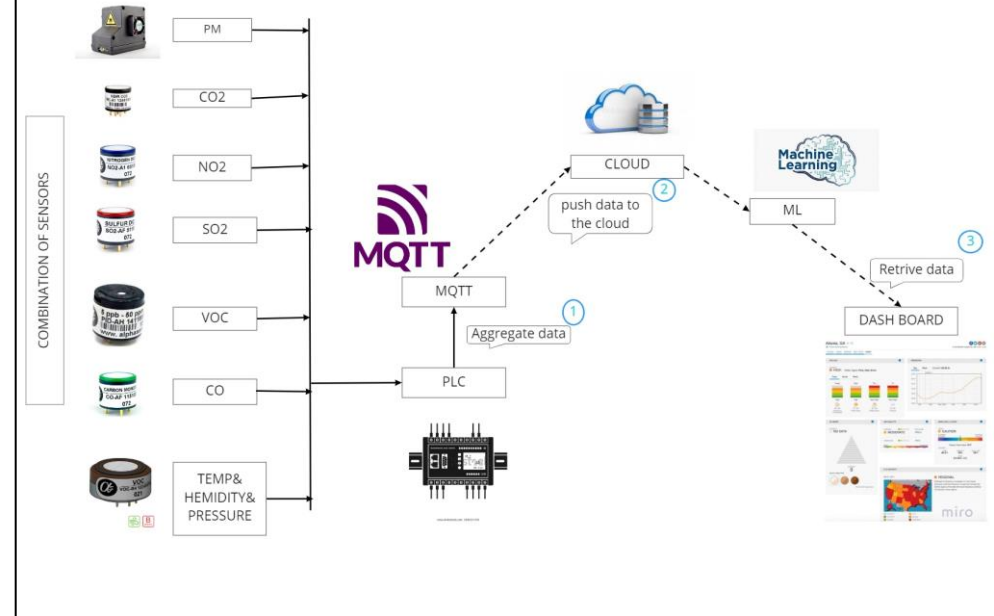
Theme Name: Port Air Quality Monitoring.

Idea/Approach Details

Solution

- Data on the air quality in the 12 Major Ports will be collected through hardware(VAAYU). Kept on the dashboard to assess the extent of measures taken to keep the air quality index within the normal level.
- In this Hardware we will use different types of sensors to monitor the air quality in various ports and put it on to a dashboard.
- The monitored Air qualities will be automatically stored in the cloud storage as data for the prediction using ML Algorithm.

FLOW CHART :



Technology Stack

- Data Collection: IOT hardware device
- Data Storage: Cloud Storage,MySQL
- ML Model: Python
- Web Application: PHP

Idea/Approach Details

Use Cases

- Air pollution causes severe damage to the environment, it has to be given much attention to keep that it into a normal level.
- Our intention is here to detect air quality in seaports using several gas sensors.

Dependencies / Show stopper

- Analysis of the **incidence** of pollutants originating in the port.
- Easy remote management of the sensor network through a **dashboard**.
- Generation of **periodic incident reports** for the Port Authority.
- **Cost Efficiency**
- **portable**



Team Member Details

Team Leader Name: Manirajan K

Branch : M.Sc

Stream : Data Analytics

Year: I

Team Member 1 Name: Sugumar S

Branch : MBA

Stream : Management

Year: I

Team Member 2 Name: Renish X

Branch : MBA

Stream : Management

Year: I

Team Member 3 Name: Ragavi K

Branch : MBA

Stream : Management

Year: I

Team Member 4 Name: Bhuvaneshwaran A

Branch : MCA

Stream : Computer Applications

Year: I

Team Member 5 Name: Vishwanath A

Branch : MCA

Stream : Computer Applications

Year: I

Team Mentor 1 Name: Dr.V. Bhuvaneshwari

Category : Academic (Associate Professor)

Expertise : Big Data Analytics, Data Analytics

Domain Experience : 16+ years

Team Mentor 2 Name: S. Saravanakumar

Category : Industry

Expertise :Data Scientist-Consultant

Domain Experience : 3+ Year